

Unifying Special Relativity and Quantum Mechanics

Lecture 16 — Gravity as a QFT

(The Massive Spin-2 Field, the Absence of Ghosts,
and the Fierz–Pauli Lagrangian)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–15: everything in the normal text and in the **yellow** / **pink** / **grey** boxes is a faithful transcription of the handwritten notes (yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones, grey boxes for the plain pencil frames). The colours of the text roughly follow the colours of the pens used in the original. Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, redrawn figures, and connections to earlier lectures.

Pagination of this scan. This lecture opens a new chapter: the page numbering restarts at 1 (the **ninth restart** in the collection of scans). The sheets are therefore marked [p. 1*****]–[p. 10*****], with **nine stars**.

One anomaly: the scan stops mid-sentence. The ten sheets are complete and in order, with no duplicates — but the lecture is *not* finished. Sheet [p. 10*****] breaks off in the middle of consideration 1) of the massless limit, at “... and cannot survive the mass—”. Section 9 transcribes the fragment as it stands and a blue box completes the sentence and sketches where the argument is going, so that the cut is not disorienting.

Where we left off. Lecture 14 built the massive spin-1 (Proca) field by imposing the absence of ghosts, and studied its massless limit; Lecture 15 showed how gauging a global $U(1)$ forces the photon’s couplings (minimal coupling). This lecture runs *exactly the same programme one spin higher*: take a symmetric rank-2 tensor $h^{\mu\nu}$, write the most general quadratic Lagrangian, and let the demand “no ghosts” fix every coefficient. The output is the Fierz–Pauli Lagrangian — the free theory of a massive graviton — and the last page starts the massless limit, where the gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ (linearized diffeomorphisms) must appear.

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Part I — The most general quadratic Lagrangian

(pp. 1*****-2*****)

1 Gravity as a QFT: the reminder and the fundamental field [p. 1*****]

We follow an approach that closely parallels what we have done in the case of electromagnetism.

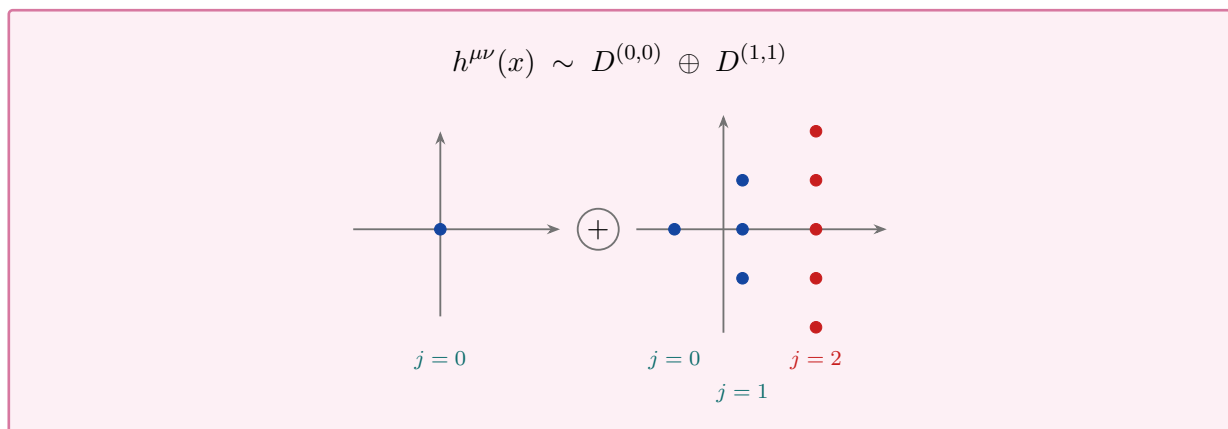
Reminder

We started from the free theory for the massive case and derived the Proca Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu, \quad \text{with E.o.M.} \quad \begin{cases} (\square + M^2)V^\mu(x) = 0 \\ \partial_\mu V^\mu(x) = 0 \end{cases}$$

We thus start from a **free theory** for the case of a **massive spin-2 particle**.

↪ We consider as **fundamental field** a **symmetric rank-2 tensor**.



↪ In the decomposition under **rotations**, $h^{\mu\nu}$ contains a $j=2$ component.

Added explanation (not in the original notes)

The counting behind the picture. A symmetric rank-2 tensor in four dimensions has $\frac{4 \cdot 5}{2} = 10$ independent components. As a representation of the Lorentz group these split into the *trace* $h = g_{\mu\nu}h^{\mu\nu}$, which is a Lorentz scalar — the $D^{(0,0)}$ — and the *traceless symmetric* part, which is the nine-dimensional $D^{(1,1)}$:

$$10 = 1 + 9.$$

Restricting to the rotation subgroup $SO(3)$ (this is the “ $\sim$ ” in the figure), the nine-dimensional piece decomposes further as

$$D^{(1,1)}|_{SO(3)} = j=0 \oplus j=1 \oplus j=2 \quad (1 + 3 + 5 = 9),$$

and this is exactly what the right-hand weight diagram shows: one isolated dot, a column of three, and a column of five. The five red dots are the $j=2$ multiplet — the states we actually want, since a **massive spin-2 particle** has $2s+1=5$ **physical polarizations**.

Why this is the whole problem in one picture. The field carries 10 components; the particle needs 5. The other 5 must be removed by constraints that follow from the equations of motion — and getting exactly the right constraints is what fixes the Lagrangian. This is the same logic as Lecture 14, where the Proca field V^μ carried 4 components and the constraint $\partial_\mu V^\mu = 0$ cut them down to the $2s+1=3$ of a massive spin-1 particle.

2 The generic Lagrangian and its six coefficients [p. 2*****]

We apply our procedure, and we start from a generic Lagrangian quadratic in $h^{\mu\nu}$ and with at most two derivatives.

$$\mathcal{L} = a(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho}) + b(\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + c(\partial_\mu h^{\mu\nu})(\partial_\nu h) + d(\partial_\mu h)(\partial^\mu h) + m^2(e h_{\mu\nu} h^{\mu\nu} + f h^2)$$

► pure kinetic terms

► “mass term”, with no derivatives

We have $h = g_{\mu\nu} h^{\mu\nu}$ with $g_{\mu\nu}$ the flat Minkowski metric.

a, b, c, d, e, f are, in principle, free coefficients. Is there a way to nail down their values?

↔ **Intuition:** the Lagrangian above is “too generic” since it describes a field $h^{\mu\nu}$ which has **10** independent components. We only want **5** of them to be physical. As in the case of the Proca Lagrangian (remember $\partial_\mu V^\mu = 0$?) we expect to have specific relations among coefficients. How to derive them?

We follow the approach discussed in the case of the Proca Lagrangian, and we impose the **absence of physical ghosts**.

Added explanation (not in the original notes)

Why exactly these five structures, and no others. The rules are: quadratic in h , at most two derivatives, Lorentz invariant, and $h^{\mu\nu}$ symmetric. With two derivatives one must contract four indices of $\partial h \partial h$; up to integration by parts (which lets you move a derivative from one factor to the other, changing a term into a combination of the others plus a total divergence) the independent contractions are exactly the four written above:

$$\partial_\mu h_{\nu\rho} \partial^\mu h^{\nu\rho}, \quad \partial_\mu h_{\nu\rho} \partial^\nu h^{\mu\rho}, \quad \partial_\mu h^{\mu\nu} \partial_\nu h, \quad \partial_\mu h \partial^\mu h.$$

With no derivatives there are only two scalars, $h_{\mu\nu} h^{\mu\nu}$ and h^2 , and the factor m^2 in front is there so that e and f are dimensionless (recall $[h] = [M]$, so $[h^2] = [M]^2$ and a mass term needs $[M]^4$).

What “physical ghost” means (Lecture 14). A ghost is a mode whose kinetic term comes with the *wrong sign*, so that its energy is unbounded from below — the theory has no stable vacuum. Equivalently, at the level of the Lagrangian, it shows up as a term with *more than two derivatives* acting on a single scalar mode: an equation of motion of order higher than two has solutions with negative-energy states (Ostrogradsky’s theorem). The strategy of the next two pages is precisely the one used for Proca: decompose $h^{\mu\nu}$ into pieces of definite spin, look at which pieces acquire higher-derivative kinetic terms, and choose the coefficients so that those terms cancel.

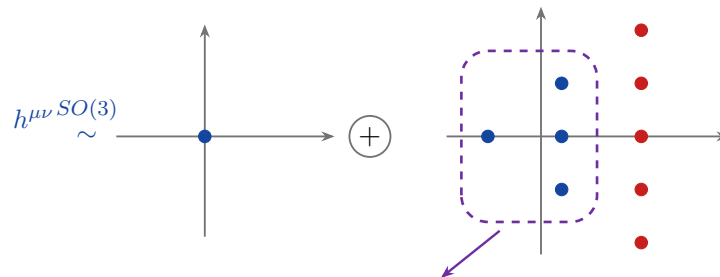
Part II — Field decomposition and the absence of ghosts (pp. 3*****-5*****)

3 Field decomposition [p. 3*****]

We need something analogue to the decomposition

$$V^\mu = V_T^\mu + (\partial^\mu \phi), \quad \partial_\mu V_T^\mu = 0.$$

We note that



- This is a 4-vector; 4 out of the 10 components of $h^{\mu\nu}$ form a 4-vector.

A 4-vector is contained into a rank-2 tensor via its derivative, so we write

$$h^{\mu\nu} = \tilde{h}^{\mu\nu} + \partial^\mu \chi^\nu + \partial^\nu \chi^\mu$$

10 components
(of $h^{\mu\nu}$)

- 4 components moved to χ^μ

$\tilde{h}^{\mu\nu}$ cannot be a generic symmetric rank-2 tensor but must satisfy 4 constraints so that it only has 6 independent components. These constraints are

$$\partial_\mu \tilde{h}^{\mu\nu} = 0$$

Added explanation (not in the original notes)

Reading the weight diagram. The four encircled blue dots are the $j = 0$ singlet of $D^{(1,1)}$ together with the $j = 1$ triplet: $1 + 3 = 4$ states, which is exactly the $SO(3)$ content of a four-vector (χ^μ has one time component, a $j = 0$, and three space components, a $j = 1$). That is the sense in which “4 of the 10 components of $h^{\mu\nu}$ form a 4-vector”.

Why the decomposition takes this shape. A four-vector cannot appear inside a rank-2 tensor on its own — the index structure does not match. The only way to build a *symmetric* rank-2 object out of χ^μ with the right dimensions is $\partial^\mu \chi^\nu + \partial^\nu \chi^\mu$. This is the direct analogue of the Proca decomposition quoted at the top: there, $V^\mu = V_T^\mu + \partial^\mu \phi$ took the scalar ϕ into a vector via one derivative; here, one derivative takes the vector χ^μ into a tensor.

The counting. $h^{\mu\nu}$ has 10 components, χ^μ takes 4 of them, so $\tilde{h}^{\mu\nu}$ must be left with $10 - 4 = 6$. But $\tilde{h}^{\mu\nu}$ is written as a symmetric rank-2 tensor, which would carry 10 — hence it must be subject to 4 conditions, and the natural Lorentz-covariant ones are the transversality conditions $\partial_\mu \tilde{h}^{\mu\nu} = 0$ (one for each free index value ν). Again the parallel with Proca is exact: there V_T^μ obeyed the single condition $\partial_\mu V_T^\mu = 0$.

A note on notation. The original writes the transverse tensor with a tilde, $\tilde{h}^{\mu\nu}$, and on the

next page introduces a second tilde for the transverse part of the vector, $\tilde{\chi}^\mu$. I have kept both tildes; they are different objects (a tensor and a vector), so no confusion arises.

4 The full decomposition and the redistribution of the 10 DOF [p. 4*****]

In turn, the four-vector χ^μ can be written as

$$\chi^\mu = \tilde{\chi}^\mu + \partial^\mu \phi, \quad \text{with} \quad \partial_\mu \tilde{\chi}^\mu = 0$$

All in all, we find the decomposition

$$h^{\mu\nu} = \tilde{h}^{\mu\nu} + \partial^\mu \tilde{\chi}^\nu + \partial^\nu \tilde{\chi}^\mu + 2 \partial^\mu \partial^\nu \phi$$

$$\partial_\mu \tilde{h}^{\mu\nu} = 0, \quad \partial_\mu \tilde{\chi}^\mu = 0$$

$$\hookrightarrow \text{The 10 DOF of } h^{\mu\nu} \text{ are “redistributed” into } \begin{cases} \tilde{h}^{\mu\nu} & 6 \text{ DOF} \\ \tilde{\chi}^\mu & 3 \text{ DOF} \\ \phi & 1 \text{ DOF} \end{cases}$$

We now substitute the above decomposition into the Lagrangian, and we impose the **absence of physical ghosts** (no terms in the Lagrangian with more than two derivatives).

One finds:

$$a = -\frac{b}{2}; \quad c = -b; \quad d = \frac{b}{2}; \quad e = -f$$

Imposing these conditions, the Lagrangian takes the form (written in terms of “ b ” and “ e ”)

Added explanation (not in the original notes)

Where the numbers 6, 3, 1 come from. $\tilde{h}^{\mu\nu}$: symmetric rank-2 (10) minus the four transversality conditions = 6. $\tilde{\chi}^\mu$: a four-vector (4) minus the one condition $\partial_\mu \tilde{\chi}^\mu = 0 = 3$. ϕ : a single scalar = 1. Total $6 + 3 + 1 = 10$, so nothing has been lost or double-counted — the decomposition is a genuine change of variables.

Why the decomposition exposes the ghost. Each derivative in the decomposition puts an extra ∂ on the lower-spin pieces: χ enters with one derivative, ϕ with *two*. When you substitute into a Lagrangian that is already quadratic in derivatives, the scalar ϕ can therefore pick up terms with up to *four* derivatives, of the schematic form $(\partial^2 \phi)(\partial^2 \phi)$. Such a term gives ϕ a fourth-order equation of motion and hence a ghost. Demanding that all such terms cancel is what produces the four relations above — one for each dangerous structure. This is the same mechanism as in Lecture 14, where the Stückelberg scalar of the Proca field had to have its higher-derivative terms cancel, forcing $\beta = -\alpha$ there.

Sanity check on the relations. They leave exactly *two* free parameters, b and e — one overall normalization for the kinetic part and one for the mass part. Both will be fixed on p. 8***** by demanding that the field satisfy the Klein–Gordon equation with the standard normalization, giving $b = -1$, $e = -\frac{1}{2}$. So of the original six coefficients, four are fixed by the ghost conditions and the remaining two by conventions — *nothing* is left free. The free massive spin-2 theory is unique.

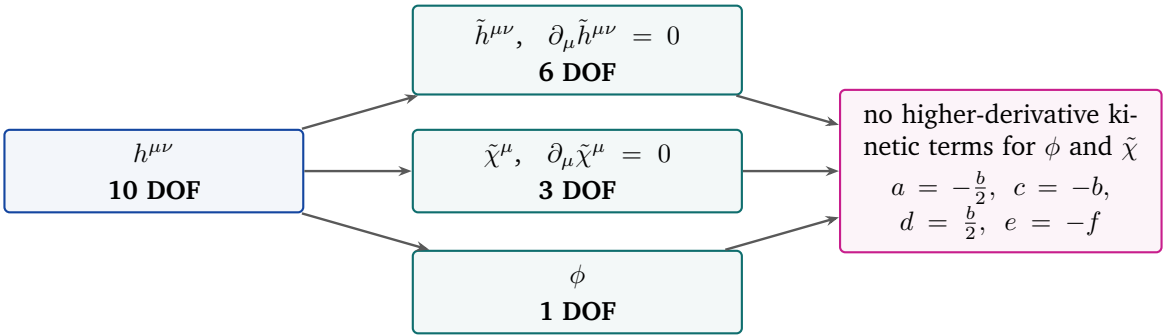


Figure 1. The logic of pp. 3*****–4*****: split the ten components by spin, then demand that the low-spin pieces carry no higher-derivative (ghostly) kinetic terms. (Added figure, not in the original notes.)

Part III — The E.o.M. and the Fierz–Pauli Lagrangian

(pp. 5*****–9*****)

5 The ghost-free Lagrangian, and the key question [p. 5*****]

$$\mathcal{L} = b \left[-\frac{1}{2} (\partial_\mu h_{\nu\rho}) (\partial^\mu h^{\nu\rho}) + (\partial_\mu h_{\nu\rho}) (\partial^\nu h^{\mu\rho}) - (\partial_\mu h^{\mu\nu}) (\partial_\nu h) \right. \\ \left. + \frac{1}{2} (\partial_\mu h) (\partial^\mu h) \right] + e m^2 (h_{\mu\nu} h^{\mu\nu} - h^2)$$

Key question: does this Lagrangian correctly propagate only the 5 DOF we are interested in?

To answer this question, we look at the E.o.M.

↔ We expect that the field $h^{\mu\nu}$ satisfies the Klein–Gordon equation and, in addition, a number of constraints that reduce the number of independent field variables from 10 to 5.

Equation of motion for $h^{\mu\nu}$

The Euler–Lagrange equations are

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu h_{\rho\sigma})} = \frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}}$$

We compute the derivatives:

■ $\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}}$ ► The only term in $\mathcal{L}$ that depends on h (no derivatives).

$$\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = \frac{\partial}{\partial h_{\rho\sigma}} \left[e m^2 (h_{\mu\nu} h^{\mu\nu} - h^2) \right] = \frac{\partial}{\partial h_{\rho\sigma}} \left[e m^2 (h_{\mu\nu} h_{\rho\sigma} g^{\mu\rho} g^{\nu\sigma} - g^{\rho\sigma} h_{\rho\sigma} h) \right]$$

[p. 6*****]

$$= 2e m^2 h_{\mu\nu} g^{\mu\rho} g^{\nu\sigma} - 2e m^2 g^{\rho\sigma} h = 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h)$$

$$\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h)$$

Added explanation (not in the original notes)

A transcription remark. In the second line above the original re-uses the letters ρ, σ both as the *free* indices of the derivative $\partial/\partial h_{\rho\sigma}$ and as the *dummy* indices being summed inside $h_{\mu\nu} h^{\mu\nu}$ and $g^{\rho\sigma} h_{\rho\sigma}$. Strictly this is not allowed — fresh dummy letters are needed — but the intended meaning is unambiguous and the result is correct. Written cleanly, the two terms are

$$\frac{\partial}{\partial h_{\rho\sigma}} (h_{\alpha\beta} h_{\gamma\delta} g^{\alpha\gamma} g^{\beta\delta}) = 2h^{\rho\sigma}, \quad \frac{\partial}{\partial h_{\rho\sigma}} (g^{\alpha\beta} h_{\alpha\beta} g^{\gamma\delta} h_{\gamma\delta}) = 2g^{\rho\sigma} h.$$

The factors of 2 arise because each quadratic expression contains the differentiated field twice.

6 The derivative term, and the equation of motion [p. 6*****]

■ $\frac{\partial \mathcal{L}}{\partial (\partial_\mu h_{\rho\sigma})}$ ► Write it as $(\partial^\nu h^{\mu\rho})$.

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\rho\sigma})} &= \frac{\partial}{\partial(\partial_\mu h_{\rho\sigma})} \left\{ b \left[-\frac{1}{2}(\partial_\mu h_{\rho\sigma})(\partial^\mu h^{\rho\sigma}) + (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) \right. \right. \\
&\quad \left. \left. - (\partial_\mu h^{\mu\nu})(\partial_\nu h) + \frac{1}{2}(\partial_\mu h)(\partial^\mu h) \right] \right\} = \\
&= b \left[-(\partial^\mu h^{\rho\sigma}) + (\partial^\rho h^{\mu\sigma}) + (\partial^\sigma h^{\mu\rho}) - g^{\mu\rho} g^{\nu\sigma} (\partial_\nu h) \right. \\
&\quad \left. - (\partial_\nu h^{\nu\mu}) g^{\rho\sigma} + g^{\rho\sigma} (\partial^\mu h) \right]
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\rho\sigma})} &= b \left[-(\partial^\mu h^{\rho\sigma}) + (\partial^\rho h^{\mu\sigma}) + (\partial^\sigma h^{\mu\rho}) - g^{\mu\rho} (\partial^\sigma h) \right. \\
&\quad \left. - (\partial_\nu h^{\nu\mu}) g^{\rho\sigma} + g^{\rho\sigma} (\partial^\mu h) \right]
\end{aligned}$$

We thus get the E.o.M. in the form [p. 7*****]

$$\begin{aligned}
b \left[-\square h^{\rho\sigma} + \partial_\mu \partial^\rho h^{\mu\sigma} + \partial_\mu \partial^\sigma h^{\mu\rho} - \partial^\rho \partial^\sigma h - \partial_\mu \partial_\nu h^{\nu\mu} g^{\rho\sigma} \right. \\
\left. + g^{\rho\sigma} \square h \right] &= 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h)
\end{aligned}$$



Added explanation (not in the original notes)

The six terms, one by one. Each term of the boxed derivative simply picks up the ∂_μ of the Euler-Lagrange equation:

$$\begin{array}{ll}
-\partial^\mu h^{\rho\sigma} & \rightarrow -\partial_\mu \partial^\mu h^{\rho\sigma} = -\square h^{\rho\sigma} \\
+\partial^\rho h^{\mu\sigma} & \rightarrow +\partial_\mu \partial^\rho h^{\mu\sigma} \\
+\partial^\sigma h^{\mu\rho} & \rightarrow +\partial_\mu \partial^\sigma h^{\mu\rho} \\
-g^{\mu\rho} \partial^\sigma h & \rightarrow -\partial^\rho \partial^\sigma h \\
-(\partial_\nu h^{\nu\mu}) g^{\rho\sigma} & \rightarrow -\partial_\mu \partial_\nu h^{\nu\mu} g^{\rho\sigma} \\
+g^{\rho\sigma} \partial^\mu h & \rightarrow +g^{\rho\sigma} \square h
\end{array}$$

The one subtlety worth knowing. In the second structure the naive differentiation gives $2 \partial^\rho h^{\mu\sigma}$, from the two factors of the term $(\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho})$; since $h^{\rho\sigma}$ is symmetric one symmetrizes the result in $(\rho\sigma)$ and writes it as $\partial^\rho h^{\mu\sigma} + \partial^\sigma h^{\mu\rho}$, which is what appears above. (The same symmetrization has *not* been performed on the fourth and fifth terms; doing so consistently would replace $-g^{\mu\rho} \partial^\sigma h$ by $-\frac{1}{2}(g^{\mu\rho} \partial^\sigma h + g^{\mu\sigma} \partial^\rho h)$ and so on. This affects nothing below: every step that follows is contracted or traced in a way that is insensitive to it.)

7 The four-divergence of the E.o.M. [p. 7*****]

Compute the four-divergence of the E.o.M. (we apply ∂_ρ)

$$b \left[-\square \partial_\rho h^{\rho\sigma} + \partial_\mu \square h^{\mu\sigma} + \partial_\mu \partial^\sigma \partial_\rho h^{\mu\rho} - \square \partial^\sigma h - \partial^\sigma \partial_\mu \partial_\nu h^{\nu\mu} + \partial^\sigma \square h \right] = 2e m^2 (\partial_\rho h^{\rho\sigma} - \partial^\sigma h)$$

The left side vanishes

We thus find $2e m^2 (\partial_\rho h^{\rho\sigma} - \partial^\sigma h) = 0$, from which we get

$$\partial_\rho h^{\rho\sigma} = \partial^\sigma h$$

We now plug in this condition back into the E.o.M. ★

$$\begin{aligned} & b \left[-\square h^{\rho\sigma} + \partial^\rho \underbrace{(\partial_\mu h^{\mu\sigma})}_{\partial^\sigma h} + \partial^\sigma \underbrace{(\partial_\mu h^{\mu\rho})}_{\partial^\rho h} - \partial^\rho \partial^\sigma h - g^{\rho\sigma} \underbrace{\partial_\mu (\partial_\nu h^{\nu\mu})}_{\partial^\mu h} + g^{\rho\sigma} \square h \right] = \\ & = b \left[-\square h^{\rho\sigma} + \partial^\rho \partial^\sigma h + \partial^\sigma \partial^\rho h - \partial^\rho \partial^\sigma h - g^{\rho\sigma} \square h + g^{\rho\sigma} \square h \right] = \\ & = b (-\square h^{\rho\sigma} + \partial^\rho \partial^\sigma h) = 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h) \end{aligned}$$

Added explanation (not in the original notes)

Why the left-hand side of the divergence vanishes. The six terms cancel in three pairs, each pair being the same expression written with the derivatives in a different order (partial derivatives commute):

$$\underbrace{-\square \partial_\rho h^{\rho\sigma}}_1 + \underbrace{\partial_\mu \square h^{\mu\sigma}}_2 = 0, \quad \underbrace{\partial^\sigma \partial_\mu \partial_\rho h^{\mu\rho}}_3 - \underbrace{\partial^\sigma \partial_\mu \partial_\nu h^{\nu\mu}}_5 = 0, \quad \underbrace{-\square \partial^\sigma h}_4 + \underbrace{\partial^\sigma \square h}_6 = 0.$$

This is not an accident: it is the statement that the *kinetic operator* of the theory is identically transverse, so its divergence carries no information. Everything that survives comes from the mass term — which is why the resulting relation is a genuine **constraint**, not a dynamical equation.

This is the spin-2 analogue of Lecture 14's $\partial_\mu V^\mu = 0$. There, taking the divergence of the Proca equation killed $\partial_\mu \partial_\nu F^{\nu\mu}$ by antisymmetry and left $M^2 \partial_\mu V^\mu = 0$. Here the same manoeuvre leaves $2em^2(\partial_\rho h^{\rho\sigma} - \partial^\sigma h) = 0$. In both cases the constraint exists *only because* $m \neq 0$ — a first hint of why the massless limit (p. 10*****) will be delicate.

8 The trace, and the Klein–Gordon equation [p. 8*****]

We thus get

$$\begin{aligned} b(-\square h^{\rho\sigma} + \partial^\rho \partial^\sigma h) &= 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h) \\ \partial_\rho h^{\rho\sigma} &= \partial^\sigma h \end{aligned}$$

★★

We take the trace (contract with $g_{\rho\sigma}$)

$$b(-\square h + \square h) = 2e m^2 (h - 4h) \rightsquigarrow 2e m^2 (-3h) = 0$$

from which we get

$$h = 0$$

Going back to ★★ and imposing this condition, we get

$$\begin{aligned} b(-\square h^{\rho\sigma} - \partial^\rho \partial^\sigma h) &= 2e m^2 (h^{\rho\sigma} - g^{\rho\sigma} h) \\ \partial_\rho h^{\rho\sigma} &= \partial^\sigma h \\ h &= 0 \end{aligned}$$

We thus write

$$\begin{aligned} -b \square h^{\rho\sigma} - 2e m^2 h^{\rho\sigma} &= 0 \\ \partial_\rho h^{\rho\sigma} &= 0 \\ h &= 0 \end{aligned}$$

↔ If we take $b = -1$ and $e = -\frac{1}{2}$, we get precisely the KG equation

$$(\square + m^2)h^{\rho\sigma} = 0$$

Added explanation (not in the original notes)

The trace step in detail. Contracting with $g_{\rho\sigma}$ uses $g_{\rho\sigma} h^{\rho\sigma} = h$, $g_{\rho\sigma} \partial^\rho \partial^\sigma = \square$ and, in four dimensions, $g_{\rho\sigma} g^{\rho\sigma} = 4$. The left-hand side gives $b(-\square h + \square h) = 0$ identically, and the right-hand side gives $2em^2(h - 4h) = -6em^2h$. Since $e \neq 0$ and $m \neq 0$ (we are describing a massive particle), the only solution is $h = 0$: **the trace is not a propagating field, it is forced to vanish.**

Then everything collapses. With $h = 0$ the earlier constraint $\partial_\rho h^{\rho\sigma} = \partial^\sigma h$ becomes $\partial_\rho h^{\rho\sigma} = 0$, and the $\partial^\rho \partial^\sigma h$ and $g^{\rho\sigma} h$ terms disappear from the equation of motion, leaving a pure Klein–Gordon operator acting on $h^{\rho\sigma}$.

Why $b = -1$, $e = -\frac{1}{2}$ is the natural choice. Substituting into $-b \square h^{\rho\sigma} - 2em^2 h^{\rho\sigma} = 0$ gives $\square h^{\rho\sigma} + m^2 h^{\rho\sigma} = 0$. Any other choice with $-b > 0$ and $-2e/b = m^2/m^2$ would rescale the field or redefine the mass; $b = -1$ fixes the canonical normalization of the kinetic term and $e = -\frac{1}{2}$ makes m the physical mass. Note that $e = -f$ from p. 4***** then gives $f = +\frac{1}{2}$: **the mass term is forced to be proportional to $h^2 - h_{\mu\nu} h^{\mu\nu}$** , with that exact relative coefficient -1 between the two structures. This is the celebrated *Fierz–Pauli tuning*: any other ratio re-introduces a ghost (the trace mode becomes dynamical with the wrong sign).

9 The Fierz–Pauli Lagrangian and the 5 DOF [p. 9*****]

With this choice of parameters, we write the Lagrangian as

Fierz–Pauli Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho}) - (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu})(\partial_\nu h) - \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{m^2}{2}(h^2 - h_{\mu\nu}h^{\mu\nu})$$

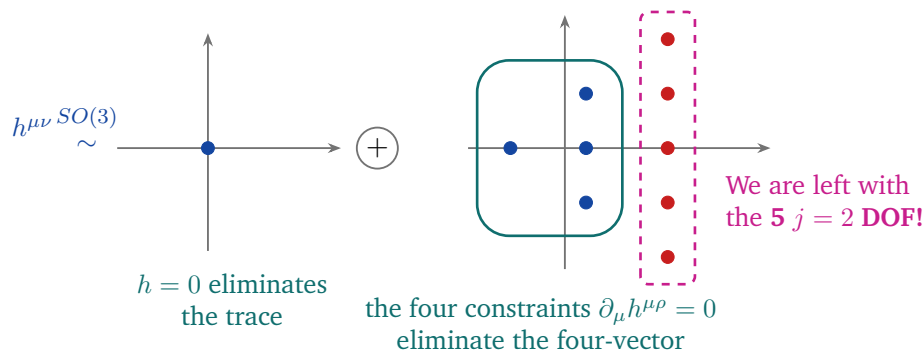
which gives the E.o.M.

$$(\square + m^2)h^{\mu\nu}(x) = 0$$

$$h = 0$$

$$\partial_\mu h^{\mu\rho} = 0$$

These are $1 + 4 = 5$ constraints, that remove 5 field components from the 10 DOF of $h^{\mu\nu}$, leaving only **5 dynamical DOF**. In more detail:



Added explanation (not in the original notes)

The counting, closed. Start from 10. The trace condition $h = 0$ removes 1; the four transversality conditions $\partial_\mu h^{\mu\rho} = 0$ remove 4; $10 - 1 - 4 = 5$. And the 5 that survive are precisely the $j = 2$ multiplet (the red column) — the physical polarizations of a massive spin-2 particle. Compare Lecture 14: for Proca, $4 - 1 = 3 = 2s + 1$ with $s = 1$. Same structure, one rung up.

What this Lagrangian is, in other words. The kinetic part of the Fierz–Pauli Lagrangian is exactly the *linearized Einstein–Hilbert action*: if one writes the spacetime metric as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and expands $\sqrt{-g} R$ to second order in h , one obtains those four kinetic structures with those relative coefficients, and nothing else. That is the sense in which this lecture is titled “gravity as a QFT”: we did not start from geometry, we started from a spin-2 particle and demanded consistency — and the linearized theory of general relativity came out.

The mass term, however, is not geometric. General relativity has no $m^2 h^2$ term: the graviton is massless. The combination $h^2 - h_{\mu\nu}h^{\mu\nu}$ found here is the unique mass term that does not propagate a ghost at quadratic order — Fierz and Pauli found it in 1939. (At *cubic* order and beyond the ghost returns for generic completions; this is the Boulware–Deser ghost, and avoiding it is what makes massive gravity hard. Well beyond this course, but worth knowing that the tuning found on p. 4***** is only the beginning of the story.)

Part IV — Towards the massless limit

(p. 10*****)

10 The massless limit [p. 10*****)

The Lagrangian is completely nailed down by imposing the absence of physical ghosts.

→ The massless limit

We are interested in the massless limit of the Fierz–Pauli Lagrangian.

Let us keep in mind the situation in the case of the spin-1 field.

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu \xrightarrow[\text{limit}]{\text{massless}} \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

↪ The Lagrangian becomes invariant under the gauge transformation $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$, as it should be to properly describe a massless spin-1 particle.

Similarly, when we take the massless limit, the field $h^{\mu\nu}(x)$ must acquire a gauge invariance under the transformation

$$h^{\mu\nu}(x) \rightarrow h^{\mu\nu}(x) + \partial^\mu \xi^\nu(x) + \partial^\nu \xi^\mu(x)$$

Two considerations:

1) The term $\frac{m^2}{2}(h^2 - h_{\mu\nu}h^{\mu\nu})$ is not invariant under a gauge transformation, and cannot survive the mass...

[The sheet, and with it the scan, breaks off here in mid-sentence.]

Added explanation (not in the original notes)

Completing the broken sentence. The sentence certainly ends “...and cannot survive the massless limit”. The point is straightforward: under $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ the two structures h^2 and $h_{\mu\nu}h^{\mu\nu}$ transform into a mess of $\partial\xi$ terms that do not cancel against each other, for any relative coefficient. A mass term is simply incompatible with gauge invariance — exactly as $\frac{1}{2}M^2V_\mu V^\mu$ was in the spin-1 case, where $V_\mu \rightarrow V_\mu + \partial_\mu \alpha$ turns it into $\frac{1}{2}M^2(V_\mu V^\mu + 2V^\mu \partial_\mu \alpha + \partial_\mu \alpha \partial^\mu \alpha)$. So the massless theory must have $m = 0$ identically, and the gauge symmetry appears only at that point.

And consideration 2), which the scan does not reach. By the parallel with Lecture 14, the second consideration is about the kinetic term: one checks that the four kinetic structures of the Fierz–Pauli Lagrangian, with exactly the coefficients found on p. 4*****, are invariant under $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ — the same tuning that removed the ghosts also produces the gauge symmetry, for free. That is a striking result, and the spin-2 counterpart of the fact that $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is automatically gauge invariant.

What ξ^μ is. Writing $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, an infinitesimal change of coordinates $x^\mu \rightarrow x^\mu - \xi^\mu(x)$ shifts the metric perturbation by exactly $\partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. So the gauge invariance of the massless spin-2 field is nothing but **invariance under infinitesimal diffeomorphisms** — general covariance, in its linearized form. The photon’s $U(1)$ has become the coordinate freedom of general relativity.

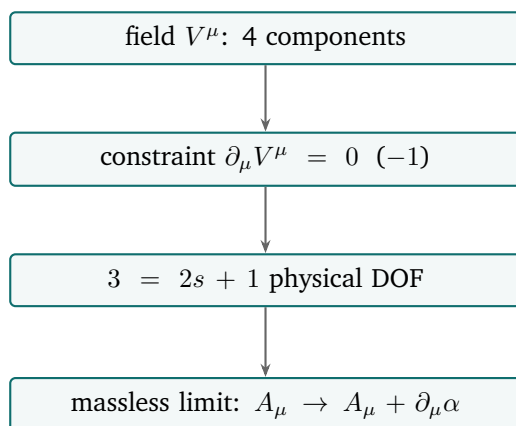
A warning about what comes next. From Lecture 14 you already know the punchline for spin-1: the massless limit is smooth provided the vector couples to a conserved current. For spin-2 it is *not*: the massive theory has 5 DOF and the massless one has 2, and the extra modes do not decouple — their coupling to the trace of the energy–momentum tensor survives $m \rightarrow 0$. The consequence is the **van Dam–Veltman–Zakharov discontinuity**: massive gravity with $m \rightarrow 0$ gives light bending different from general relativity by a finite amount, no matter

how small m is. (This was flagged in a blue box in Lecture 14; the next lecture presumably develops it.)

Summary of Lecture 16 — the logic in seven lines

1. A massive spin-2 particle needs $2s + 1 = 5$ polarizations; the natural field is a symmetric rank-2 tensor $h^{\mu\nu}$, which carries 10 components. Five of them must be removed.
2. Write the most general Lagrangian: quadratic in h , at most two derivatives. It has six free coefficients $a, \dots, f$.
3. Decompose $h^{\mu\nu} = \tilde{h}^{\mu\nu} + \partial^\mu \tilde{\chi}^\nu + \partial^\nu \tilde{\chi}^\mu + 2\partial^\mu \partial^\nu \phi$ (6 + 3 + 1 DOF) and demand that no piece acquire a higher-derivative — i.e. ghostly — kinetic term. This fixes four relations: $a = -\frac{b}{2}, c = -b, d = \frac{b}{2}, e = -f$.
4. The equation of motion, contracted with ∂_ρ , gives the constraint $\partial_\rho h^{\rho\sigma} = \partial^\sigma h$; contracted with $g_{\rho\sigma}$ it gives $h = 0$.
5. Feeding both back in leaves $(\square + m^2)h^{\mu\nu} = 0$ once $b = -1$ and $e = -\frac{1}{2}$ — so the last two coefficients are fixed too.
6. The result is the **Fierz–Pauli Lagrangian**, with 1 + 4 = 5 constraints leaving exactly the 5 physical $j = 2$ degrees of freedom. Its kinetic part is the linearized Einstein–Hilbert action.
7. In the massless limit the field must acquire the gauge invariance $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$ — linearized diffeomorphisms. The mass term cannot survive it. *(The scan stops here.)*

Spin 1 (Lecture 14)



Spin 2 (this lecture)

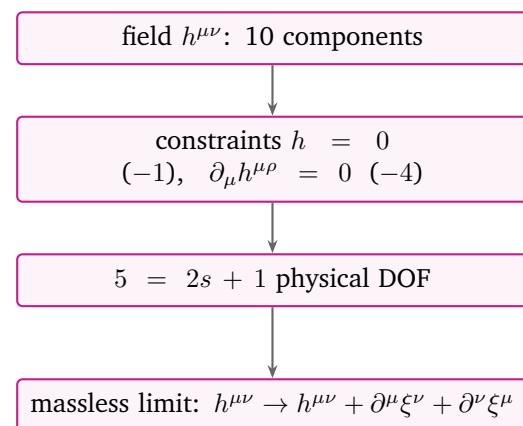


Figure 2. The two constructions side by side. Every step of this lecture has an exact counterpart one spin lower. *(Added figure, not in the original notes.)*